

A 120-GHz Class-F Frequency Doubler With 7.8-dBm P_{OUT} in 55-nm Bulk CMOS

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Abstract—This article analyzes the 2nd-order harmonic power generation with power level boosted by fundamental and 3rd-order harmonic based on the Krummenacher-Vittoz (EKV) models. The optimal waveforms of fundamental and 3rd-order harmonic are derived for maximizing the 2nd-order harmonic power. Adopting this method, a class-F frequency doubler in a 55-nm bulk CMOS technology is designed, fabricated, and measured. The doubler prototype achieves a 3-dB operation bandwidth of 112–125 GHz, a maximum output power (P_{OUT}) of 7.8 dBm, a maximum conversion gain (CG) of -0.46 dB, a maximum efficiency of 6.8% with dc power consumption of 88 mW, and a core area of only 0.13 mm². To the best of our knowledge, the proposed class-F doubler achieves the highest P_{OUT} among similar CMOS works and is comparable to those in advanced SiGe technologies.

Index Terms—Class-F, CMOS, Enz-Krummenacher-Vittoz (EKV) model, frequency doubler, harmonic generation, push-push, submillimeter, terahertz.

I. INTRODUCTION

SILICON-BASED high-resolution imaging, high-speed point-to-point (P2P) wireless communication, and short-range frequency-modulated continuous wave (FMCW) radar in millimeter-wave (mm-wave) frequencies have been proven to yield significant potential in future advanced applications [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12]. Yet, challenges remain due to the limited high-frequency performance of silicon-based devices, especially in the frequency source circuits, which is the essential module in the systems. A practical solution is to upconvert high-quality low-frequency signal sources to high-frequency using frequency multiplication.

In mm-wave systems, the frequency doublers are widely used for their simple topology, easy harmonic extraction, and

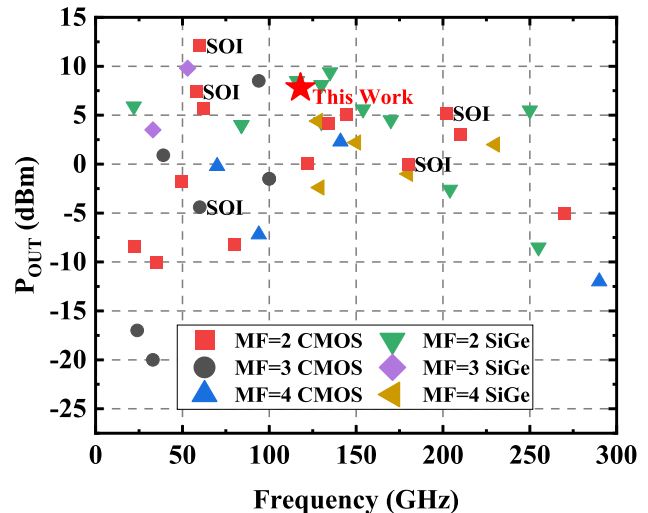


Fig. 1. Summary of state-of-the-art mm-wave and terahertz frequency multiplier (20–300 GHz) in CMOS and SiGe technologies. (MF = 2: doubler; MF = 3: tripler; and MF = 4: quadrupler) [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [35], [36], [37], [42], [43], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59].

high output power compared to frequency multipliers with higher multiplication factors, as plotted in Fig. 1. Typically, the output power of the doubler must reach the optimal operation point of the next-stage circuits, e.g., the mixers and power amplifiers. Due to the high loss characteristics of passive circuits and the f_{MAX} limitations of active circuits, the output power of silicon-based high-frequency doublers, especially in a CMOS process, is less than 5 dBm at mm-wave [14], [15], [16], [17], [18], [19], [20].

The output power of a frequency multiplier is studied and enhanced by two means: 1) recover nontarget harmonics [21] and 2) maximize the nonlinearity of devices to generate extra target harmonics [23]. To analyze the nonlinearity of devices, the harmonic balance (HB) method is one of the common methods. However, the HB method is usually time-consuming and not intuitive, which is suitable only for simulations using EDA tools. In the literature [22], HB analysis was used to disassemble the device into a superposition of linear and nonlinear networks to obtain the optimal amplitude and phase conditions to the maximization fundamental output power of the device, and this conclusion was also applied to the literature [24] for a VCO close to f_{MAX} . Volterra–Wiener theory was used for the study of nonlinear systems in [25]. By manipulating the device nonlinearity and optimizing the

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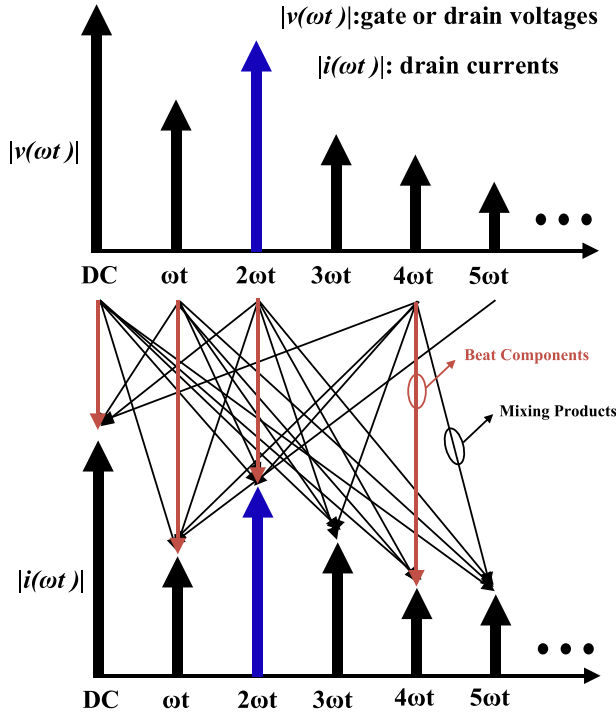


Fig. 2. Frequency-domain mappings of a nonlinear FET. $|v(\omega t)|$ represents the gate or drain voltages, and $|i(\omega t)|$ represents the drain currents.

embedding network, optimum conditions for harmonic power generation are provided. In [26] and [27], the nonlinearity generated by the transistor transforming between different operation regions under large signal excitations based on a time-varying behavior model was analyzed, while the relationship between the interconversion of each harmonic was obtained, as shown in Fig. 2 [30].

In this article, we analyze the harmonic power generation of transistors based on the Enz–Krummenacher–Vittoz (EKV) models and derive the relationship of fundamental and 3rd-order harmonic onto the 2nd-order harmonic power generation. The boosting or weakening conditions of the fundamental and 3rd-order harmonic for the 2nd-order harmonic power generation are obtained analytically. Based on the theoretical analysis, we propose a class-F frequency doubler to realize the optimal fundamental and 3rd-order harmonic to maximize the 2nd-order harmonic power generation.

This article is organized as follows. Section II analyzes the 2nd-order harmonic power generation based on the EKV model. Section III finds the optimal phase conditions and waveforms for the 2nd-order harmonic power generation. Section IV performs the HB simulation validation between the theory and physical model. In Section V, a 120-GHz class-F frequency doubler prototype is proposed to boost the 2nd-order harmonic power generation. Section VI demonstrates the simulation and measurement results of the proposed class-F frequency doubler. Section VII provides a summary of the results.

II. SECOND HARMONIC POWER GENERATION

It is assumed that the input excitation at the transistor gate terminal is described as $v_G = V_0 + |V_G| \sin(\omega t)$, which

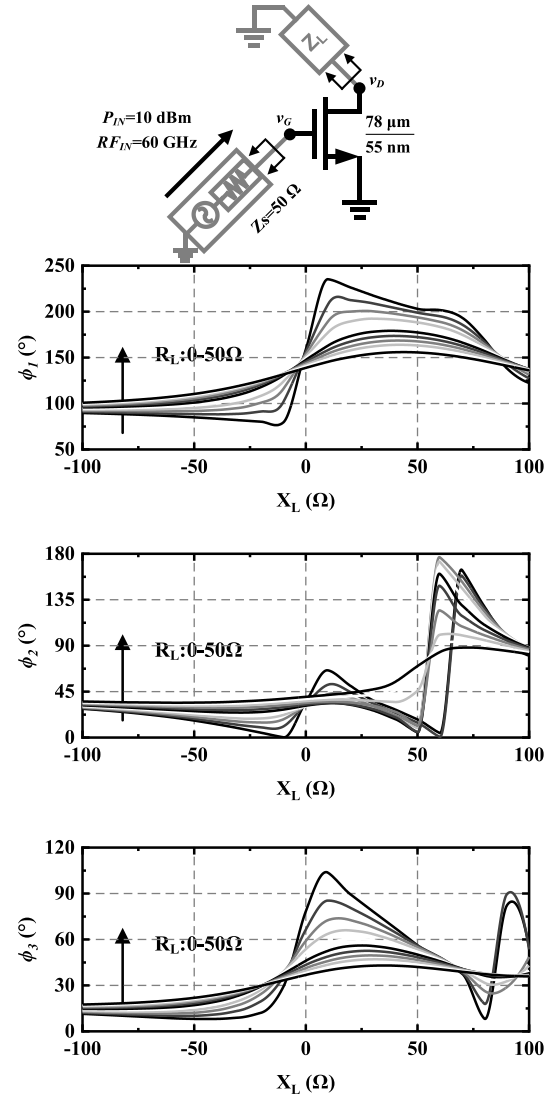


Fig. 3. Simulated ϕ_n modified by load impedance of a 78 $\mu\text{m}/55\text{ nm}$ in 55-nm bulk CMOS technology.

resembles a nonlinear memory system [33]. V_0 is the dc biasing voltage at the gate terminal and $|V_G|$ represents the fundamental voltage amplitude. By the nature of a CMOS transistor, its drain voltage v_D must contain rich harmonic components in the following expression:

$$\begin{aligned} v_D = & V_{DD} + |V_{D1}| \sin(\omega t + \phi_1) + |V_{D2}| \sin(2\omega t + 2\phi_2) \\ & + |V_{D3}| \sin(3\omega t + 3\phi_3) \\ & + \dots + |V_{Dn}| \sin(n \cdot \omega t + 3\phi_n) \end{aligned} \quad (1)$$

where ϕ_n is the relative phase shift of each harmonic voltage using the phase of v_G as a reference according to Groszkowski's assumptions [40], [41]. Fig. 3 shows that ϕ_n is modified by the load impedance of the transistor. Actually, ϕ_n is also modified by ω , $|V_G|$, and input impedance Z_S , which makes ϕ_n flexible and versatile, and each term in ϕ_n is nonlinear with each other.

For a conventional push–push doubler [34], the drain terminal of pseudo-differential pairs (PDPs) is connected directly to

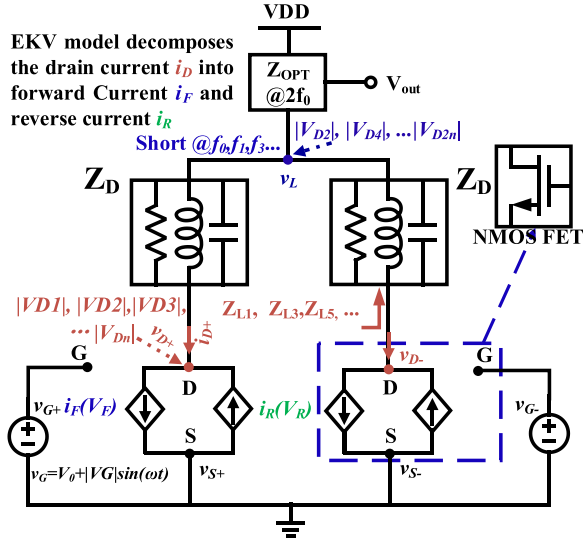


Fig. 4. EKV model of the proposed class-F doubler topology.

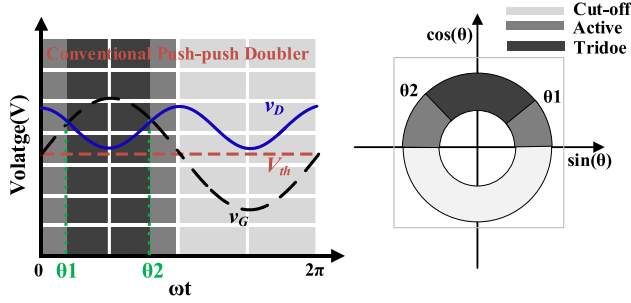


Fig. 5. Observation of operation regions in a conventional push-push doubler: gate and drain terminal waveforms in large signal and OM of half-circuit.

cancel out its odd harmonics, i.e., $|V_{D1,3,5,\dots,n}| = 0$, and boost the 2nd-order harmonic. Theoretically, in a nonlinear system, each harmonic produces other harmonics or the so-called beat components or mixed products [28], [29], [30].

In Fig. 4, the proposed class-F doubler topology imitates a square v_D waveform after modulating the values of ϕ_n , so that the odd harmonics are also capable of enhancing the 2nd-order harmonic power. Here, the EKV model deconstructs the NMOS FET [31], [32]. The critical difference between a class-F doubler and the conventional push-push topology is that the transistor drain terminal retains the odd harmonics due to the presence of Z_D , and the imaginary ground of the odd harmonics is moved to behind Z_D . Moreover, the odd harmonic impedance by looking out from the transistor is now no longer 0, but $Z_{L1,3,5,\dots,n}$ determined by Z_D . For the EKV model, the drain current is decomposed into a forward current i_F dependent on forward voltage V_F and reverse current i_R dependent on reverse voltage V_R . i_F and i_R are fictitious currents inside the FET, and their superposition constitutes i_D at the drain terminal. Assuming square-law MOSFET I - V characteristic, then i_F and i_R are defined by $SLR(\cdot)$, the square-

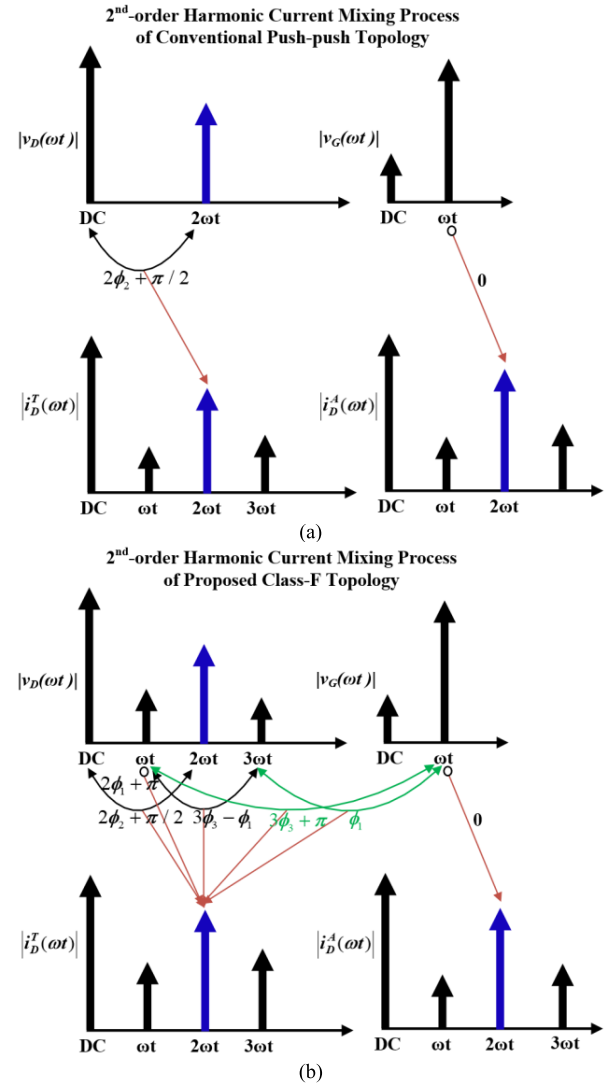
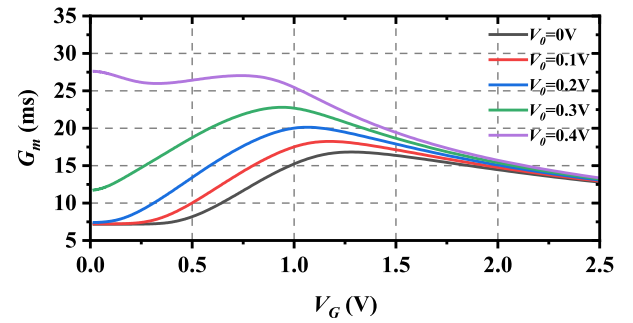


Fig. 6. Mixing process of the 2nd-order harmonic currents and the phases distinction. (a) In conventional push-push topology. (b) In proposed class-F topology.

Fig. 7. Large signal transconductance using HB simulation for a 68- μm /55-nm FET in 55-nm bulk CMOS technology with different bias voltages V_0 .

law rectifier function [33]

$$i_{F,R} = \frac{\beta}{2} \text{SLR}(V_{F,R}) \quad (2)$$

where $\text{SLR}(x) = x^2$ for $x > 0$, otherwise is 0, and β is the transconductance of the transistor when the transistor source terminal is grounded, and V_F and V_R are defined as follows:

$$V_F = v_G - v_S - V_{th}, \quad V_R = v_G - v_D - V_{th}. \quad (3)$$

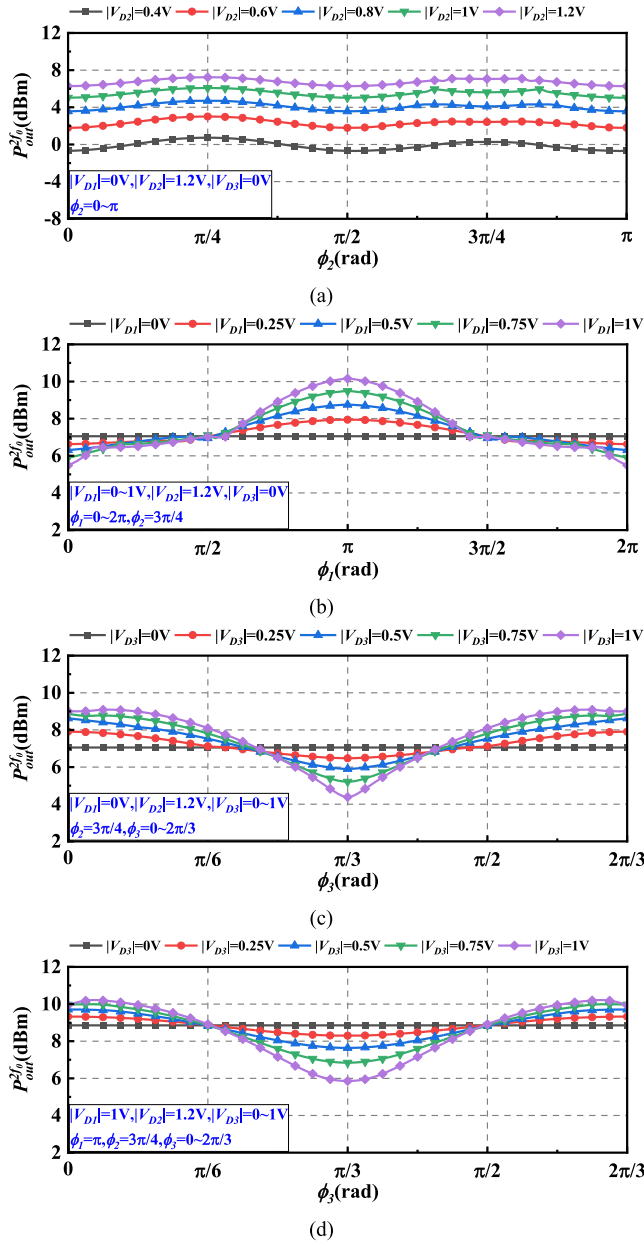


Fig. 8. Calculation of $P_{out}^{2f_0}$ using EKV model of a 68- $\mu\text{m}/55\text{-nm}$ FET in 55-nm bulk CMOS technology. (a) $|V_{D2}|$ and ϕ_2 vary from 0.4 to 1.2 V and $0-\pi$ when $|V_{D1,3}| = 0$ V. (b) $|V_{D1}|$ and ϕ_1 vary from 0 to 1 V and $0-2\pi$ when $|V_{D2}| = 1.2$ V, $\phi_2 = 3\pi/4$, and $|V_{D3}| = 0$ V. (c) $|V_{D3}|$ and ϕ_3 vary from 0 to 1 V and $0-2\pi/3$ when $|V_{D1}| = 0$ V, $|V_{D2}| = 1.2$ V, and $\phi_2 = 3\pi/4$. (d) $|V_{D3}|$ and ϕ_3 vary from 0 to 1 V and $0-2\pi/3$ when $\phi_1 = 0$, $|V_{D2}| = 1.2$ V, $\phi_2 = 3\pi/4$, and $|V_{D3}| = 1$ V.

Then, in different transistor operation regions, i_D is expressed as: in active region $i_D = i_F$, in triode region $i_D = i_F - i_R$, and in cutoff region $i_D = 0$.

In large signal, transistors cycle between different operation regions leading to nonlinearity generation [27]. Fig. 5 shows the large-signal voltage waveform at gate and drain terminals in a conventional push–push doubler and the operation map (OM), which visually illustrate transitions between cutoff, active, and triode regions [26]. When $v_G - V_{th} = v_D$, the

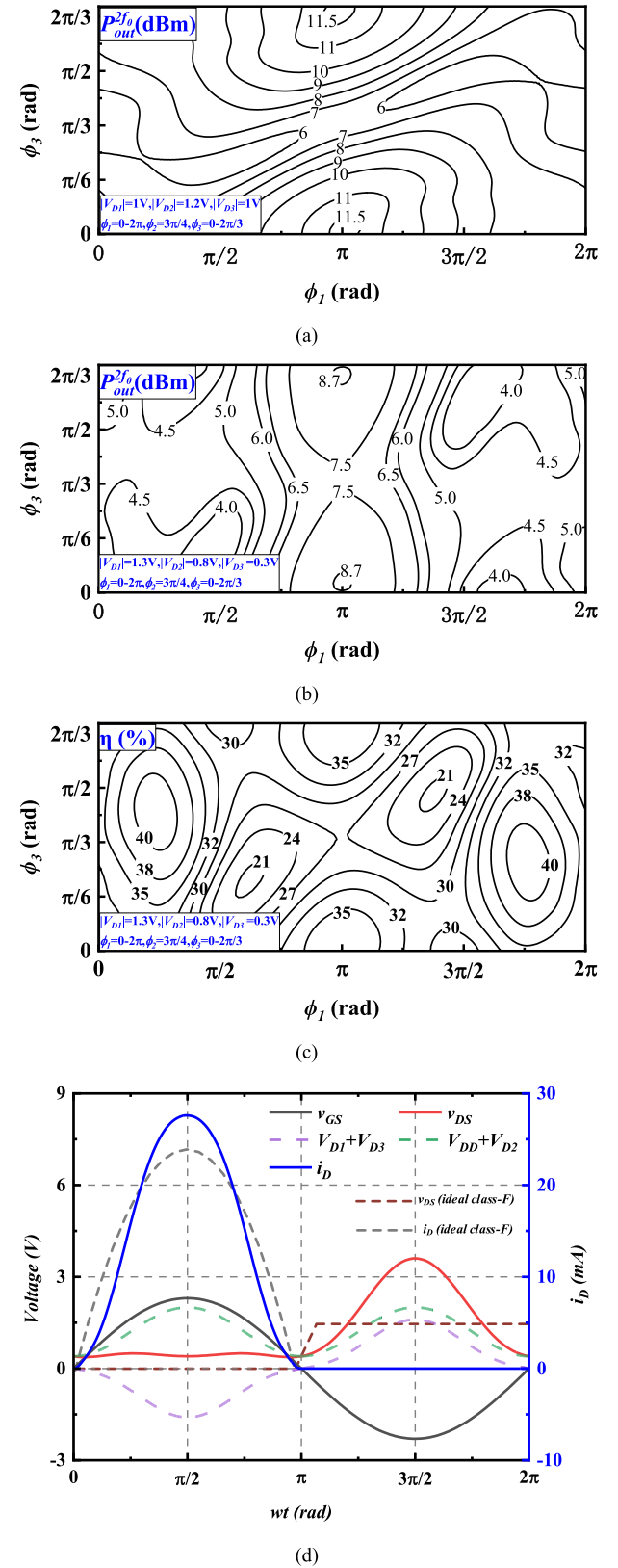


Fig. 9. Contours of the variation of $P_{out}^{2f_0}$ and η varying from ϕ_1 and ϕ_3 and the optimal waveform of v_{DS} and i_D when $|V_{D1}|$ and $|V_{D3}|$ coexist. (a) $|V_{D1}| = 1.2$ V, $|V_{D2}| = 1.2$ V, $\phi_2 = 3\pi/4$, and $|V_{D3}| = 1$ V. (b) $|V_{D1}| = 1.3$ V, $|V_{D2}| = 0.8$ V, $\phi_2 = 3\pi/4$, and $|V_{D3}| = 0.3$ V. (c) $|V_{D1}| = 1.3$ V, $|V_{D2}| = 0.8$ V, $\phi_2 = 3\pi/4$, and $|V_{D3}| = 0.3$ V. (d) Optimal waveform of v_{DS} when $|V_{D1}| = 1.3$ V, $|V_{D2}| = 0.8$ V, $\phi_2 = 3\pi/4$, and $|V_{D3}| = 0.3$ V and the ideal class-F waveform.

boundary points of the active and triode regions are defined as θ_1 and θ_2 , respectively.

Conventional doublers focus more on the conduction angle optimization, which is usually $\leq 90^\circ$ for maximizing the 2nd-order harmonic, i.e., class-B/-C doubler. Nevertheless, this article emphasizes the transistor drain terminal waveform in active and triode regions. Therefore, for the convenience of the analysis, the bias voltage V_0 is assumed equal to the threshold voltage V_{th} .

The 2nd-order harmonic power generation in the active and triode regions of the transistors can be calculated separately based on the EKV model. Only the fundamental, 2nd-, and 3rd-order harmonics are calculated since the higher harmonics are negligible.

Substitute (2) into (3), the 2nd-order harmonic drain terminal currents in the active region i_{D2}^A and triode region i_{D2}^T are derived in (4) and (5), respectively. Compared to i_{D2}^A , i_{D2}^T clearly has more harmonic compositions. In the active region, the 2nd-order harmonic current is determined only by the gate terminal input voltage amplitude $|V_G|$ and the 2nd-order term of drain terminal voltage amplitude $|V_{D2}|$. While in the triode region, the 2nd-order harmonic current is a mixed superposition of all harmonics. Fig. 6 shows the mixing process of the 2nd-order harmonic current generation in conventional push-push topology and proposed class-F topology doublers. In Fig. 6(a), $|V_{D1}|$ and $|V_{D3}|$ are eliminated for conventional push-push configuration, while in Fig. 6(b), the presence of $|V_{D1}|$ and $|V_{D3}|$ introduces more mixing terms with distinct phases

$$i_{D2}^A = -\beta \frac{|V_G|^2}{4} \cos(2\omega t) \quad (4)$$

$$i_{D2}^T = -\frac{1}{4}\beta \begin{bmatrix} 2|V_G||V_{D1}|\cos(2\omega t + \phi_1) \\ +|V_{D1}|^2\cos(2\omega t + 2\phi_1 + \pi) \\ +4V_{DD}|V_{D2}|\cos(2\omega t + 2\phi_2 + \pi/2) \\ +2|V_{D1}||V_{D3}|\cos(2\omega t + 3\phi_3 - \phi_1) \\ +2|V_G||V_{D3}|\cos(2\omega t + 3\phi_3 + \pi) \end{bmatrix} \quad (5)$$

The 2nd-order harmonic power generation $P_{out}^{2f_0}$ is divided into power generation in the active region $P_{out,A}^{2f_0}$ and power generation in the triode region $P_{out,T}^{2f_0}$. The boundary points θ_1 and θ_2 are derived by the following equation:

$$|V_G|\sin\theta_{1,2} = V_{DD} + |V_{D1}|\sin(\theta_{1,2} + \phi_1)$$

$$+ |V_{D2}|\sin(2\theta_{1,2} + 2\phi_2) \\ + |V_{D3}|\sin(3\theta_{1,2} + 3\phi_3). \quad (6)$$

In the active region, the 2nd-order harmonic power generation $P_{out,A}^{2f_0}$ is expressed in (7), as shown at the bottom of the page. Intuitively, $P_{out,A}^{2f_0}$ seems to be unrelated to $|V_{D1}|$ and $|V_{D3}|$. However, θ_1 and θ_2 are determined by (6), which contains $|V_{D1}|$ and $|V_{D3}|$. In the triode region, the 2nd-order harmonic power generation $P_{out,T}^{2f_0}$ is expressed in (8), as shown at the bottom of the page. Actually, the terms $P_{out,T}^{2f_0}$ result from the mixing between v_G and v_D . The phases depending on ϕ_n , θ_1 , and θ_2 of each sub-term in $P_{out,A}^{2f_0}$ and $P_{out,T}^{2f_0}$ are different, so the effect of their phase differences is not negligible for the 2nd-order harmonic power generation.

The $P_{out}^{2f_0}$ in one cycle now is obtained from the following equation, which is the superposition of the 2nd-order harmonic power generation in the different transistor operation regions

$$P_{out}^{2f_0} = \left| P_{out,A}^{2f_0} + P_{out,T}^{2f_0} \right| \\ = \left| \frac{1}{2\pi} \left(\int_0^{\theta_1} i_{D2}^A v_{D2} + \int_{\theta_1}^{\theta_2} i_{D2}^T v_{D2} + \int_{\theta_2}^{\pi} i_{D2}^A v_{D2} \right) \right|. \quad (9)$$

It is clear that the class-F topology produces rich nonlinearity for the odd harmonics causing by Z_D at the drain terminal. However, whether the additional nonlinearity enhances or weakens the 2nd-order harmonic power generation depends on the amplitude components $|V_{Dn}|$ and the phase shift components ϕ_n . Therefore, optimizing $|V_{Dn}|$ and ϕ_n is necessary for reshaping the transistor drain terminal voltage and current waveforms in the active and triode regions and the optimal waveform for maximizing the 2nd-order harmonic power generation.

To further simplify $P_{out}^{2f_0}$, $P_{out,A}^{2f_0}$ is decomposed into $P1$ and $P2$ and $P_{out,T}^{2f_0}$ is decomposed into $P3$ and $P4$ as derived in Appendix A. The maximum amplitude of $P1$ – $P4$ derived in Appendix A shows that $|P1|_{\max} \ll |P2|_{\max}$ and $|P3|_{\max} \ll |P4|_{\max}$, and the maximum values of $|P2|$ and $|P4|$ are dependent on the size of the triode region ($\theta_2 - \theta_1$). With $(\theta_2 - \theta_1)$ increasing, $P4$ will increase, while $P2$ will decrease. Cause $|P2|_{\max} \ll |P4|_{\max}$ when $|V_{D1}|$ and $|V_{D3}|$ are at the large signal. Therefore, it is expected to maximize $P4$ for maximizing $P_{out}^{2f_0}$ and the simplified $P_{out,sim}^{2f_0}$ is derived in (10). To maximize $P4$, (11) shows the optimal ϕ_{1-3} . Meanwhile, for larger $P4$, the triode region ($\theta_2 - \theta_1$) is better, and so is the $|V_{Dn}|$. Then, the priority goal is to tune the phases of the

$$P_{out,A}^{2f_0} = -\frac{\beta|V_G|^2|V_{D2}|}{32} [(4\pi + 4\theta_1 - 4\theta_2)\sin 2\phi_2 - \cos 2\phi_2 + \cos(4\theta_2 + 2\phi_2) - \cos(4\theta_1 + 2\phi_2)] \quad (7)$$

$$P_{out,T}^{2f_0} = -\frac{\beta V_{DD}}{32} \left\{ \begin{aligned} &2|V_G||V_{D1}|\cos(4\theta_1 + \phi_1 + 2\phi_2) - \cos(4\theta_2 + \phi_1 + 2\phi_2) \\ &-|V_{D1}|^2[\cos(4\theta_1 + 2\phi_1 + 2\phi_2) - \cos(4\theta_2 + 2\phi_1 + 2\phi_2)] \\ &+ 4V_{DD}|V_{D2}|\sin(4\theta_1 + 4\phi_2) - \sin(4\theta_2 + 4\phi_2) \\ &+ 2|V_{D1}||V_{D3}|\cos(4\theta_1 - \phi_1 + 2\phi_2 + 3\phi_3) - \cos(4\theta_2 - \phi_1 + 2\phi_2 + 3\phi_3) \\ &- 2|V_G||V_{D3}|\cos(4\theta_1 + 2\phi_2 + 3\phi_3) - \cos(4\theta_2 + 2\phi_2 + 3\phi_3) \\ &+ (\theta_1 - \theta_2) \left[\begin{aligned} &8|V_G||V_{D1}|\sin(\phi_1 - 2\phi_2) - 4|V_{D1}|^2\sin(2\phi_1 - 2\phi_2) \\ &+ 8|V_G||V_{D3}|\sin(2\phi_2 - 3\phi_3) - 8|V_{D1}||V_{D3}|\sin(\phi_1 + 2\phi_2 - 3\phi_3) \\ &- 16|V_{D2}|V_{DD} \end{aligned} \right] \end{aligned} \right\}. \quad (8)$$

fundamental, 2nd, and 3rd harmonics of v_D into the correct regions that enhance the 2nd-harmonic power generation as derived in (10) and (11), as shown at the bottom of the page.

III. OPTIMAL WAVEFORM DERIVATION

In the previous derivation, the expression for $P_{\text{out}}^{2f_0}$ of a single transistor in the class-F topology was obtained based on the EKV model in (10). After simplification, the optimal ϕ_n for the maximum $P_{\text{out}}^{2f_0}$ when V_{D1} and V_{D3} are introduced, and the magnitude of V_{Dn} also shows that the larger the better when the optimal phase condition is satisfied. This section will quantify the specific effects of $|V_{Dn}|$ and ϕ_n in four cases.

A transistor in 55-nm bulk CMOS technology is an example for simulation verifications. The β value of the transistor is replaced by the effective transconductance G_m considering the effect of compression under large signals [33]. Fig. 7 shows that the transistor G_m will gradually compress as $|V_G|$ increases. Note that the values of $|V_G|$, $|V_{Dn}|$, and ϕ_n are pre-determined in this section regardless of whether they can be implemented in an actual circuit. $V_G = 2.5$ V and $V_0 = V_{\text{th}} = 0.3$ V in the following calculations.

Fig. 8(a)–(d) shows $P_{\text{out}}^{2f_0}$ in the four cases.

- 1) Fig. 8(a), a special case that v_D containing only the 2nd-order harmonic component degenerating as a conventional push–push topology situation, shows that $P_{\text{out}}^{2f_0}$ reaches maximum when $\phi_2 = \pi/4$ or $3\pi/4$.
- 2) Fig. 8(b), the simultaneous presence of fundamental and 2nd-order harmonic to find the optimal ϕ_1 with $\phi_2 = 3\pi/4$ [which is optimal from (10)]. The engagement of fundamental in v_D shows that $P_{\text{out}}^{2f_0}$ is deteriorated when $\phi_1 = 0$ or 2π and boosted when $\phi_1 = \pi$.
- 3) Fig. 8(c), the simultaneous presence of 2nd- and 3rd-order harmonics, is illustrated to locate the optimal ϕ_3 with $\phi_2 = 3\pi/4$. It is clear that $P_{\text{out}}^{2f_0}$ is significantly improved by the 3rd-order harmonic and the optimal $\phi_3 = 0$ or $2\pi/3$. Conversely, when $\phi_3 = \pi/3$, $P_{\text{out}}^{2f_0}$ is deteriorated.
- 4) Fig. 8(d), the fundamental, 2nd-, and 3rd-order harmonics are present simultaneously, which is closer to the real circuits. $P_{\text{out}}^{2f_0}$ can be further boosted compared with cases (b) and (c).

To summarize, tuning V_{D1} alone shows a 3.1-dB boost of $P_{\text{out}}^{2f_0}$ when $|V_{D1}| = 1$ V, as shown in Fig. 8(b), tuning V_{D3} alone shows a 2.1-dB boost of $P_{\text{out}}^{2f_0}$ when $|V_{D3}| = 1$ V, as shown in Fig. 8(c), and tuning $V_{D1,3}$ shows a boost of 4.6 dB when $|V_{D1}| = 1$ V and $|V_{D3}| = 1$ V, as shown in Fig. 8(d).

Fig. 9(a) and (b) shows the contours of the variation of $P_{\text{out}}^{2f_0}$ varying from $\phi_{1,3}$ with $\phi_2 = 3\pi/4$ when $|V_G| = 2.5$ V, $|V_{D1}| = 1$ V, $|V_{D2}| = 1.2$ V, and $|V_{D3}| = 1$ V [case (d)] and when $|V_G| = 2.3$ V, $|V_{D1}| = 1.3$ V, $|V_{D2}| = 0.8$ V, and $|V_{D3}| = 0.3$ V (from the realized value of the proposed class-F doubler). Fig. 9(c) shows the contours of the variation of drain efficiency η varying from $\phi_{1,3}$ with $\phi_2 = 3\pi/4$ when $|V_G| = 2.3$ V, $|V_{D1}| = 1.3$ V, $|V_{D2}| = 0.8$ V, and $|V_{D3}| = 0.3$ V. It is clear that when the fundamental, 2nd- and 3rd-order harmonics are present simultaneously, the optimal values of ϕ_1 and ϕ_3 are not the same as when the fundamental or 3rd-order harmonics are introduced alone [cases (b) and (c)]. This is because the phase of each term in (10) is the result of the combination of $\theta_{1,2}\theta_2$ and ϕ_n . Nevertheless, both Figs. 8(d) and 9(a) and (b) demonstrate that the existence of odd harmonics at the drain terminal boosts the $P_{\text{out}}^{2f_0}$ with proper ϕ_n . Compared Fig. 9(b) and (c), the optimum condition for maximizing η is $\phi_1 \approx 0$ or 2π and $\phi_3 = \pi/3$, while the optimum condition for maximizing $P_{\text{out}}^{2f_0}$ is $\phi_1 = \pi$ and $\phi_3 \approx 0$ or $2\pi/3$, which still leads to a very high efficiency from Fig. 9(c). Thus, the proposed topology ensures simultaneous high output power and efficiency, compared to standard topology without harmonic optimization. The derivation of η is explained in Appendix B.

The optimal waveform of v_{DS} and i_D when $|V_G| = 2.3$ V, $|V_{D1}| = 1.3$ V, $|V_{D2}| = 0.8$ V, $\phi_2 = 3\pi/4$, and $|V_{D3}| = 0.3$ V and ideal class-F waveform are shown in Fig. 9(c). Compared to the ideal class-F waveform, the proposed class-F doubler topology can imitate a half-square v_{DS} and a half-sine of i_D to boost $P_{\text{out}}^{2f_0}$ with proper ϕ_n .

From the derivations and the calculation validations, the optimal waveform of class-F doubler for maximizing $P_{\text{out}}^{2f_0}$ is shown in (12). The phase conditions of V_{D1-3} make the peak of $V_{DD} + V_{D2}$ overlap the valley of $V_{D1} + V_{D3}$, which manipulated the waveform of v_D is similar to a class-F square waveform [44], [45], [46]

$$v_D = V_{DD} + |V_{D1}| \sin(\omega t + \pi) + |V_{D2}| \sin\left(2\omega t + \frac{3}{2}\pi\right) + |V_{D3}| \sin(3\omega t). \quad (12)$$

IV. HB VALIDATIONS

In Fig. 10(a), the harmonic impedances are optimized separately for individual control of each harmonic. It is analyzed in a half differential circuit. The source impedance is pre-optimized to $Z_{\text{Sopt}} = 1 + j \times 25$ for a 68- $\mu\text{m}/55\text{-nm}$ FET

$$\begin{aligned} P_{\text{out,sim}}^{2f_0} &= -\frac{\beta|V_{D2}|}{32} [4(\pi - (\theta_2 - \theta_1)) \sin 2\phi_2] |V_G|^2 \\ &\quad - \frac{\beta|V_{D2}|}{32} \left\{ (\theta_2 - \theta_1) \left[16|V_{D2}|V_{DD} - 8|V_G||V_{D1}| \sin(\phi_1 - 2\phi_2) + 4|V_{D1}|^2 \sin(2\phi_1 - 2\phi_2) \right] \right. \\ &\quad \left. - 8|V_G||V_{D3}| \sin(2\phi_2 - 3\phi_3) + 8|V_{D1}||V_{D3}| \sin(\phi_1 + 2\phi_2 - 3\phi_3) \right\} \\ &= -\frac{\beta|V_{D2}|}{32} (P2) - \frac{\beta|V_{D2}|}{32} (P4) \end{aligned} \quad (10)$$

$$\phi_1 = \pi \pm 2k\pi, \quad \phi_2 = \frac{3\pi}{4} \pm k\pi, \quad \phi_3 = 0 \pm \frac{2}{3}k\pi, \quad (k = \pm N) \quad (11)$$

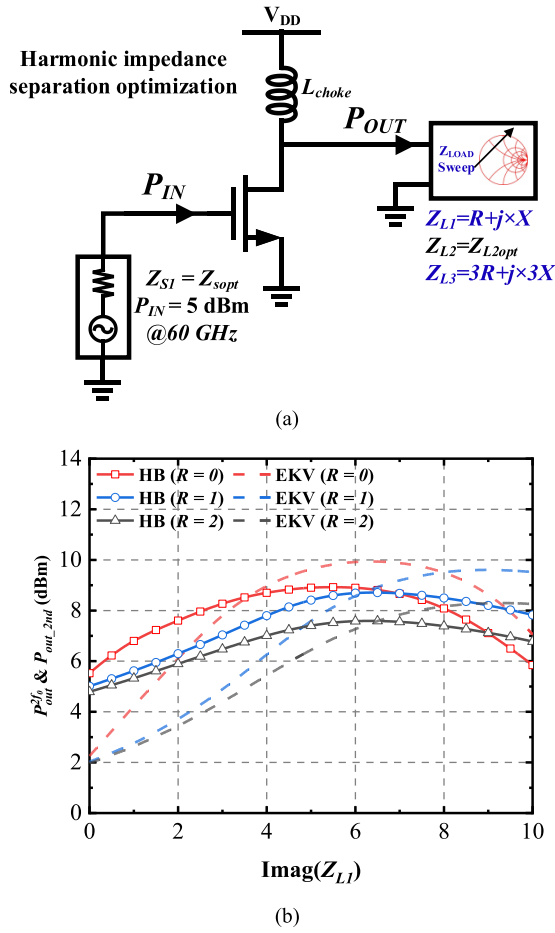


Fig. 10. Comparison of the $P_{out}^{2f_0}$ calculated based on the EKV model and $P_{out,2nd}$ simulated by the behavioral model. (a) Harmonic impedance separation optimization schematic. (b) HB simulation results and theoretical calculation results.

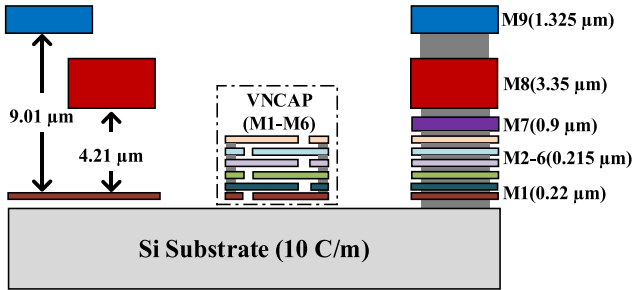


Fig. 11. GF 55-nm CMOS technology cross-sectional view.

in 55-nm bulk CMOS technology, and the fundamental, 2nd, and 3rd-order harmonics of the load impedance are set, respectively. In the PDPs, the fundamental and 3rd-order harmonics form virtual ground at the junction, so Z_{L1} and Z_{L3} are chosen to contain only imaginary parts when the load impedance of the fundamental and 3rd-order harmonic is assumed lossless. Fig. 10(b) shows the comparison of the 2nd-order harmonic generation power $P_{out}^{2f_0}$ in EKV model and the 2nd-order harmonic output power $P_{out,2nd}$ based on HB simulation in the behavioral model, when losses are added for nonzero real-part impedance as $Z_{L1} = R + j \times X$ and $Z_{L3} = 3R + j \times 3X$.

The theory still fits well with the simulation, for reasonable $R = 1-2$ in the typical silicon processes at 120 GHz.

Fig. 10 shows the comparison of the theoretical calculation based on EKV model results and HB simulation results with the variation of R and X . It is noted that the calculated $P_{out}^{2f_0}$ fits well with the simulated, and both the calculated and simulated results demonstrate that the introduction of the fundamental and 3rd harmonics at the drain terminal boosts the $P_{out}^{2f_0}$ at proper X values. The difference between calculation and simulation comes from the fact that only the fundamental, 2nd-, and 3rd-order harmonics are considered in the calculation, while the simulation takes into account the higher order harmonic components.

V. 120-GHz CLASS-F DOUBLER DESIGN

f_T and f_{max} of the 55-nm CMOS technology used in this design are 205 and 280 GHz. There are nine metal layers, as shown in Fig. 11, including three thick metal layers M7, M8, and M9, for inductor implementation. The material of M1–M8 is copper (Cu), and the material of M9 is aluminum (Al).

A prototype design of a class-F doubler is shown in Fig. 12. The input signal at f_0 is converted into a differential signal through the input matching network (IMN) for feeding into the PDP. Then, harmonics are generated by nonlinear devices M1 and M2, and then, each harmonic is filtered and shaped by the output matching network (OMN). Finally, the 2nd-order harmonic $2f_0$ reaches the output. The passive components are custom-designed using full-wave EM simulator HFSS, with the foundry-provided BEOL stack information.

The IMN consists of a low k transformer-based balun, an open slot coplanar waveguide (CPW) TL1 = 156°, and a short CPW TL2 = 9.5°. The input balun uses two top thick metal layers M8 and M9 for a better quality factor (Q). The turns ratio of the primary coil to the secondary coil is about 1:2, with a primary coil inductance $L_P = 71 \text{ pH}$ and a secondary coil inductance $L_S = 126 \text{ pH}$. The coupling coefficient $k = 0.35$ is for higher $|V_G|$, although some bandwidth is compromised. The center-tap capacitance $C_1 = 37 \text{ fF}$ is used to compensate for the phase and amplitude unbalance of the balun [34] and the resistance $R_1 = 8 \text{ k}\Omega$ is used to block the RF signal.

TL1 is used to eliminate the 2nd-order harmonic signal leaking from the drain terminal to the gate terminal. This leaking 2nd-order harmonic signal regenerates the 2nd-order harmonic current, which is out phase with the original 2nd-order harmonic current and will lead to a decrease for $P_{out}^{2f_0}$. TL1 also ensures that almost only the fundamental exists at the transistor gate terminal, which is consistent with the theoretically derived hypothesis. TL2 is a short CPW and is used to adjust the input matching.

The OMN contains the CPW TL3, TL4, bypass capacitor C_2 , the matching capacitor C_3 , and the parasitic capacitor C_{PAD} of the PAD. TL3 is used to form an odd harmonic impedance Z_{L1} and Z_{L3} at the drain terminal; then, $|V_{D1}|$ and $|V_{D3}|$ can be generated. Meanwhile, ϕ_1 and ϕ_3 are also regulated by TL3.

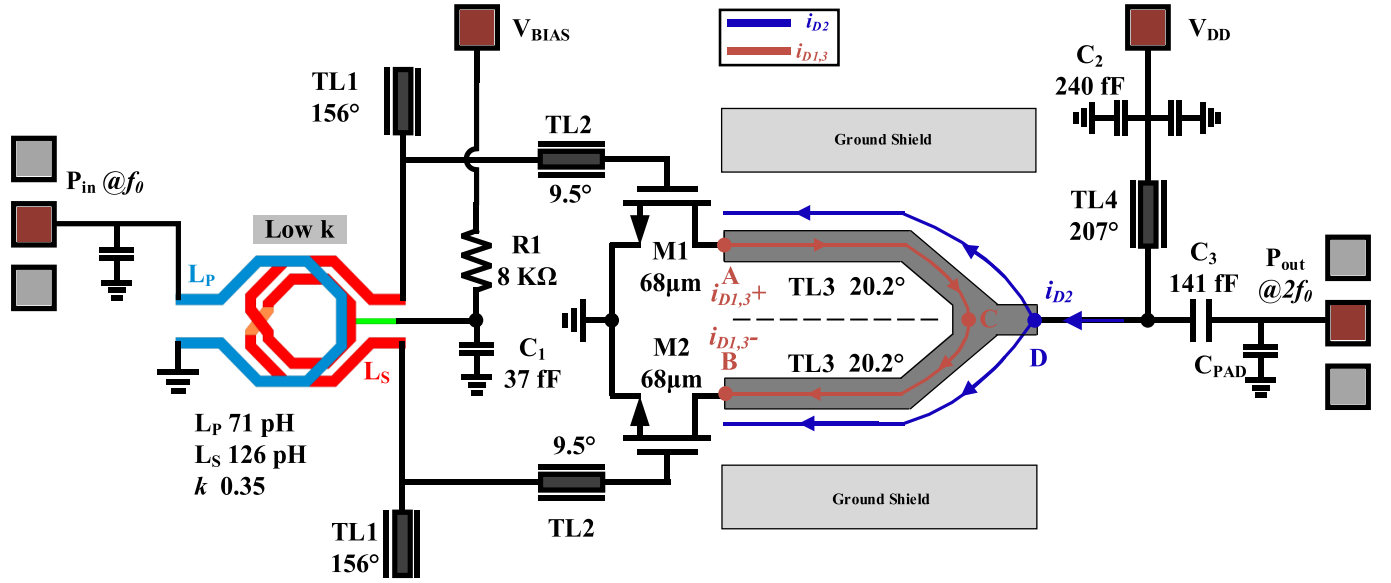


Fig. 12. Schematic of the proposed class-F doubler.

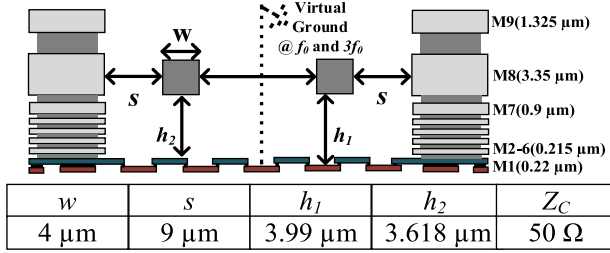


Fig. 13. Layout configuration of TL3.

To obtain the optimal Z_{L1} and Z_{L3} , TL3 needs to be carefully designed. Fig. 13 shows the layout configuration of TL3. One side of TL3 is the ground shield, which is built from metal M1–M9, and the other side is virtual ground at the odd harmonic. The odd harmonic currents i_{D1} and i_{D3} flow from node A into node B and form a virtual ground at node C. The 2nd-order harmonic current i_{D2} is divided into two parts at node D and flows to A and B, respectively.

The width w of TL3 is 4 μm , the spacing to ground shield s is 9 μm , and the distances to the bottom grid ground h_1 and h_2 are 3.99 and 3.618 μm , respectively. The characteristic impedance and propagation constants for even and odd modes are simulated by electromagnetic field analysis, as shown in Fig. 14. The odd-mode characteristic impedances $Z_{C\text{odd}}$ at 60 and 180 GHz are 50.2 and 49.5 Ω , and the even-mode characteristic impedance $Z_{C\text{even}}$ at 120 GHz is 85.8 Ω . The propagation constants β_{Si} are 2832, 4947, and 7803 rad/m at 60, 120, and 180 GHz, respectively. Suppose the length of TL3 is L_3 , then Z_{L1} and Z_{L3} can be derived by (13), where $\beta_{\text{Si_odd}}$ is the odd mode propagation constant in silica and θ_{odd} is the electrical length of TL3 at odd mode

$$Z_{L\text{odd}} = jZ_{C\text{odd}} \tan \beta_{\text{Si_odd}} L_3 = jZ_{C\text{odd}} \tan \theta_{\text{odd}}. \quad (13)$$

The HB schematic simulation results of the proposed class-F doubler with different L_3 values are shown in Fig. 15. With

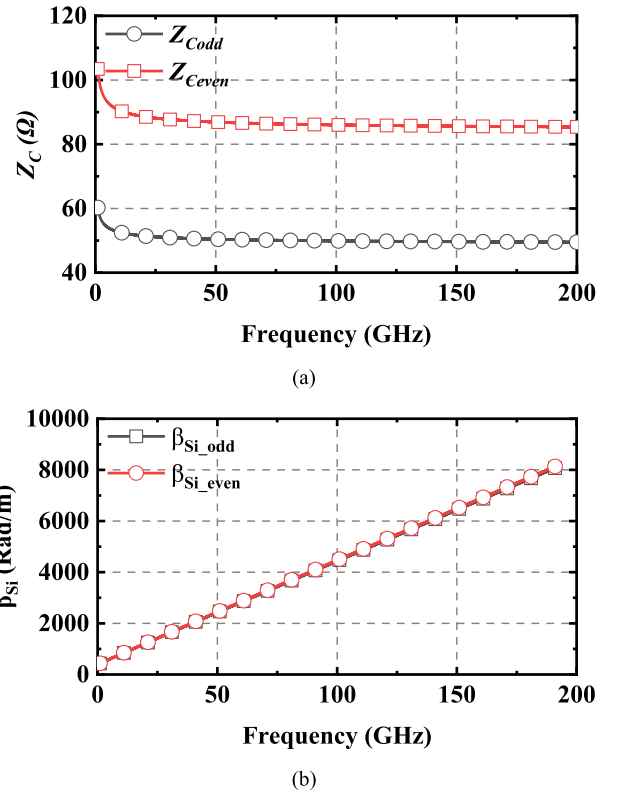


Fig. 14. EM Simulation results of TL3. (a) Characteristic impedance. (b) Propagation constants.

the increase of L_3 , P_{OUT} will increase first and then decrease, and the optimal L_3 is 120 μm for maximum P_{OUT} . Using (13), the optimal $Z_{L1} = j \times 14.0$ and $Z_{L3} = j \times 53.5$. For typical TL in CMOS at 120 GHz, the input impedance Z_{in} from EM simulation of TL3 is shown in Fig. 15(b). Here, $Z_{\text{in}} = 1 + j \times 14.7$ and $Z_{\text{in}} = 3.2 + j \times 55.5$ are obtained when the frequency is at fundamental and 3rd-order harmonic, respectively. Thus, R is around 1, and the proposed design still has advantages compared to the conventional design.

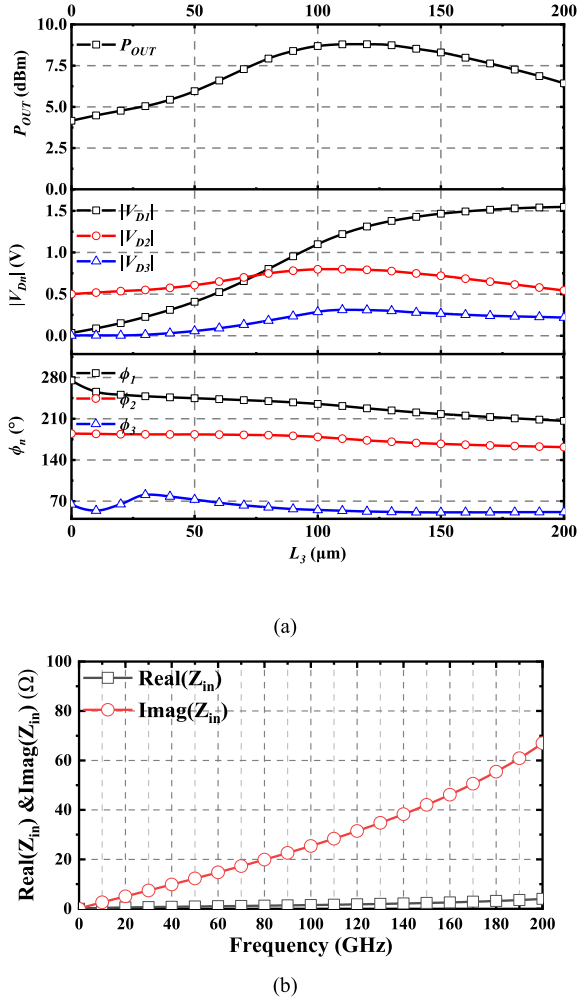


Fig. 15. (a) Simulated results with different TL3 lengths L_3 vary from 0 to 200 μm . (b) Simulated input impedance of TL3 implemented in layout with a length of 120 μm when the end of TL3 is short to ground.

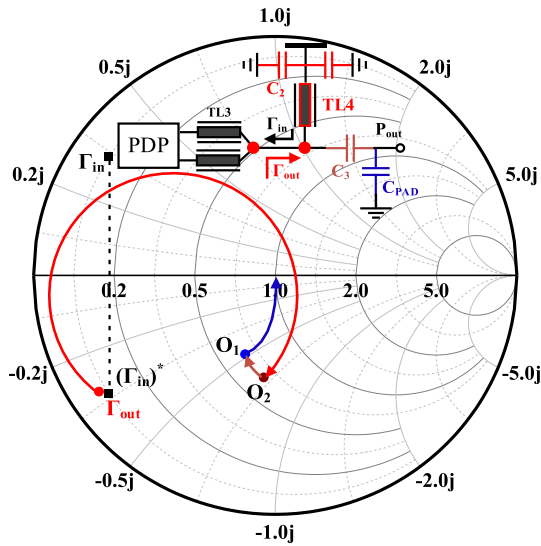


Fig. 16. Synthesized impedances Γ_{out} and OMN transforming Γ_{out} to 50 Ω .

Then, the procedure for output matching is shown in Fig. 16. The first step is to optimize the length of TL3 for the fundamental and 3rd-order harmonics tuning. Then, the PDP

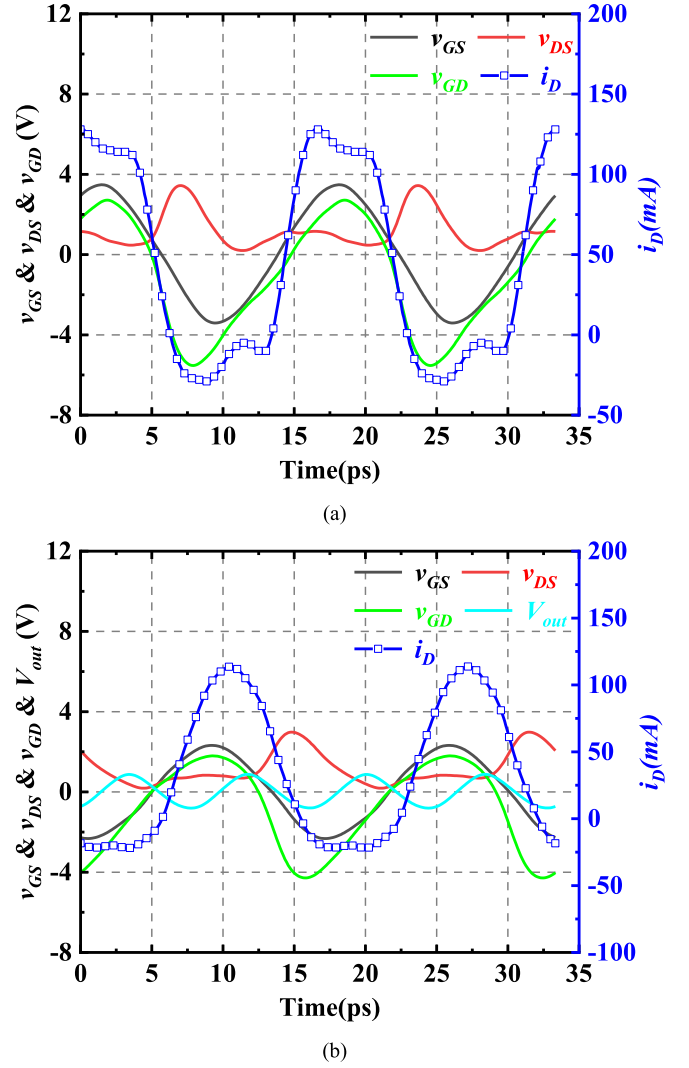


Fig. 17. (a) Pre-simulated time domain waveforms of the v_{GS} , v_{DS} , and i_D of M1 in the proposed class-F doubler schematic. (b) Post-simulated time-domain waveforms of the v_{GS} , v_{DS} , and i_D of M1 and the output voltage V_{out} in the proposed class-F doubler layout.

with TL3 and ideal optimal input impedance are combined to calculate the input reflection coefficient Γ_{in} . For maximum power transfer, Γ_{in} and the output reflection coefficient Γ_{out} are expected to be conjugate. Starting from the 50- Ω load at the output, the parasitic capacitance of the output PAD is also included in the OMN, where C_{PAD} converts the 50 Ω to the O_1 point. Then, capacitor C_3 , which is both for matching impedance and isolating dc current, converts the impedance from O_1 to the O_2 point, and then, the impedance is converted from the O_2 point to Γ_{out} by TL4 and C_2 . The reason why Γ_{out} is not exactly the same as $(\Gamma_{in})^*$ is that TL4 and C_2 need to match and isolate the output signal leakage to the power supply as well.

Fig. 17 shows the comparison of pre-simulated and post-simulated waveforms of the gate terminal voltage, drain terminal voltage, drain terminal current of M1, and the output voltage V_{out} in the proposed class-F doubler prototype. For the pre-simulation of the circuit, as shown in Fig. 17(a), the v_{DS}

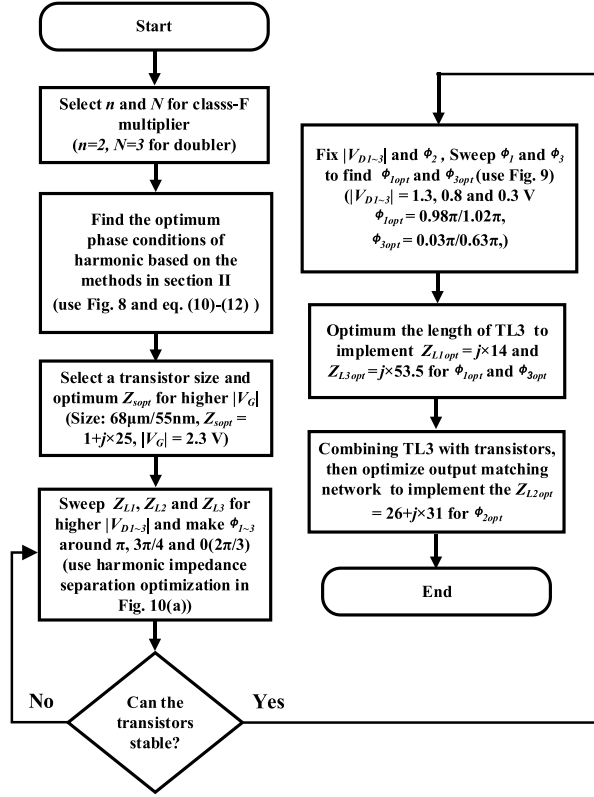


Fig. 18. Design procedure to find the optimum harmonics at the drain terminals of the transistors in a class-F doubler for maximum 2nd-harmonic output power generation. This method can be used for any n -order frequency multiplier with N -order harmonic tuning.

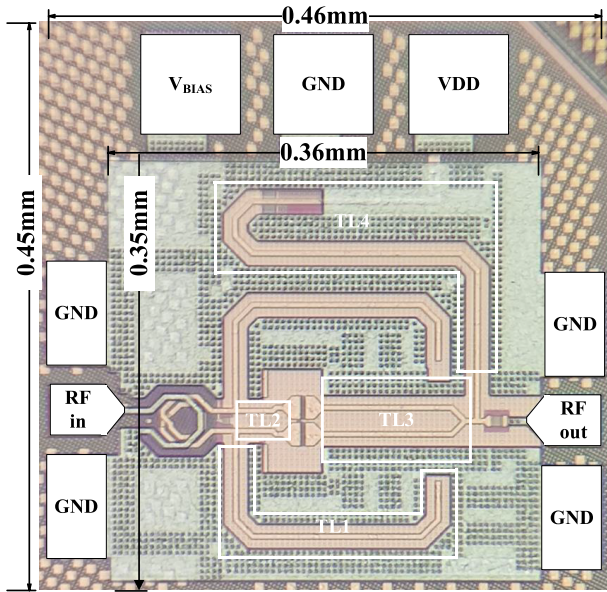


Fig. 19. Microphotograph of the proposed class-F doubler.

waveform is a square-like waveform, while the peak of the current waveform i_D overlaps the valley of v_{DS} . While in the post-simulation of the circuit, the waveform is slightly shifted by the parasitic parameters of the transistor, the nonoptimal V_{Dn} tuning, and the parasitic capacitance of TL3. In addition,

higher harmonics are included in the practical circuit, which has some modifications to the waveforms of i_D and v_{DS} . Nevertheless, it still shows that the simulated waveforms fit well with the optimal waveform based on the EKV model in Fig. 9(d), and the gate terminal voltage illustrates that the 2nd-order harmonic leakage from drain terminal to gate terminal is almost none. The drain terminal voltage nearly imitates a half-square waveform in the triode region, and the drain terminal current is shaped as a half-sine waveform, which is close to the ideal class-F waveform.

Fig. 18 shows the design procedure of the proposed class-F doubler. Here, supposing $n = 2$ and $N = 3$ for our design, where n and N represent the n -order frequency multiplier with N -order harmonic tuning. The first step is to find the optimum phase conditions using Fig. 8 and (10)–(12). Then, the optimum transistor size with IMN for largest $|V_G|$ needs to be found, and then to find the higher $|V_{D1-3}|$, the harmonic impedance separation optimization as shown in revised Fig. 10(a) is used to sweep Z_{L1-3} . In this step, $Z_{L1,3}$ and Z_{L2} are supposed to be independent because of the intrinsic properties of the class-F doubler topology. After getting high enough $|V_{D1-3}|$ which confirms the transistor is stable and the optimum $\phi_2 = 3\pi/4$, ϕ_1 and ϕ_3 are swept to find ϕ_{1opt} and ϕ_{3opt} for optimum waveform, as shown in Fig. 9. Then, the transmission line TL3 is optimum to implement Z_{L1opt} and Z_{L3opt} . At last, combining TL3 with the PDP, the OMN is optimized for Z_{L2opt} . This method can be used for any n -order frequency multiplier with N -order harmonic tuning.

VI. MEASUREMENT RESULTS

The proposed class-F doubler prototype is fabricated in a 55-nm bulk CMOS technology. The microphotograph of the chip is presented in Fig. 19. The chip area is 0.46×0.45 mm, including bond pads and the core area is 0.36×0.35 mm.

The performance of the proposed class-F doubler is measured using the platform shown in Fig. 20. The modules and connections used in the proposed class-F measurement setup are shown in Fig. 20(a). The RF input signal (53–66 GHz) is generated by the R&S SMA 100B linked to the Infinity I67 probe and fed to the device under test (DUT). The insertion loss from the input signal source to the I67 GSG probe tip is 6.7 dB at 60 GHz. The output signal generated by the chip is exported through the Infinity I140 probe, followed by WR8 Bend. The insertion loss from the I140 GSG probe tip to the end of WR8 Bend is 4.2 dB at 120 GHz. For frequency measurement, WR8 Bend is connected to the R&S FS Z140 sub-harmonic mixer, which receives the LO signal from the R&S FSUP50 spectrum analyzer and downmixes the received 90–140-GHz signal to get the IF signal returning to the spectrum analyzer, so that the output frequency of the chip can be obtained. For power testing, the W8 Bend needs to be connected to the WR8-WR10 Taper first, and then, the output power is received in Erickson Power Meter (PM5). Note that the parasitic capacitance of the input and output GSG PADs of the doubler is part of the matching networks is crucial. Consequently, their losses are not extracted from the measurement.

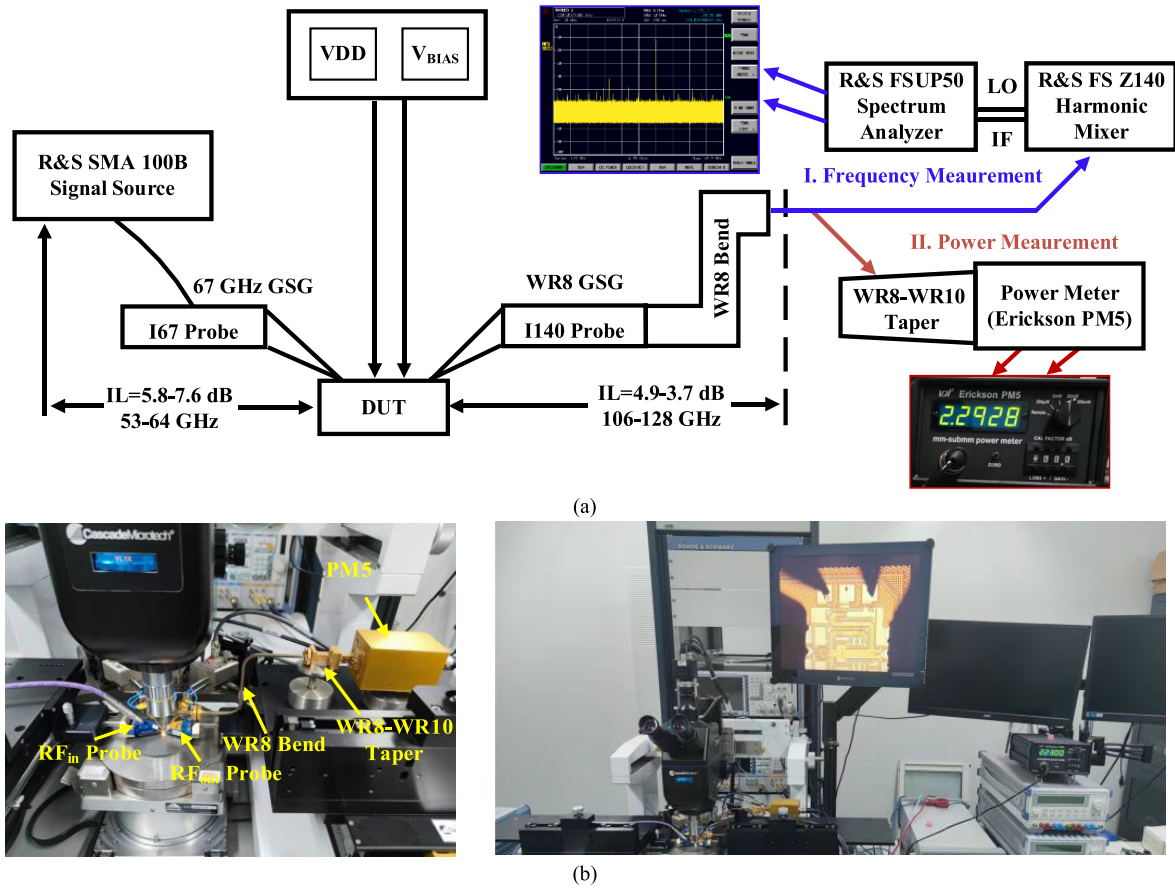


Fig. 20. Measurement setup of the proposed class-F doubler. (a) Block diagram displaying different components used in the measurement setup. (b) Photograph of the in-house measurement setup.

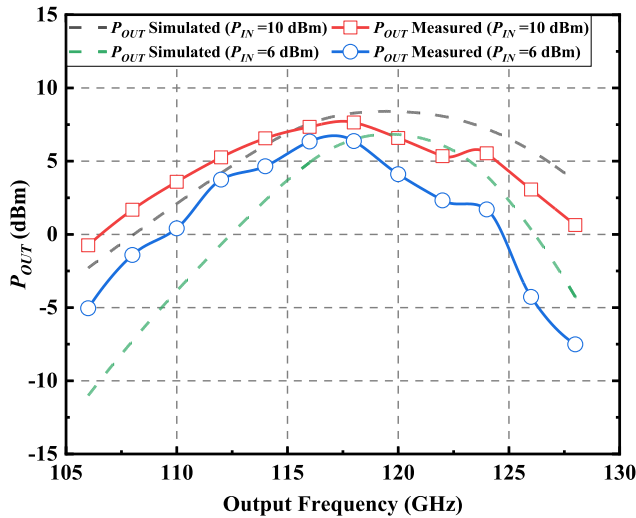


Fig. 21. Measured and simulated P_{OUT} versus P_{IN} .

The measured class-F doubler used a standard supply voltage of 1.2 V and a bias voltage of 0.3 V. Under such supply conditions, the maximum dc current was 73.2 mA, leading to a dc power consumption of 88 mW. To guarantee that the input power delivered to the chip matches the intended value, the input chain is calibrated using a 67-GHz network analyzer to calibrate the IL of 5.8–7.6 dB at 53–64 GHz. To guarantee

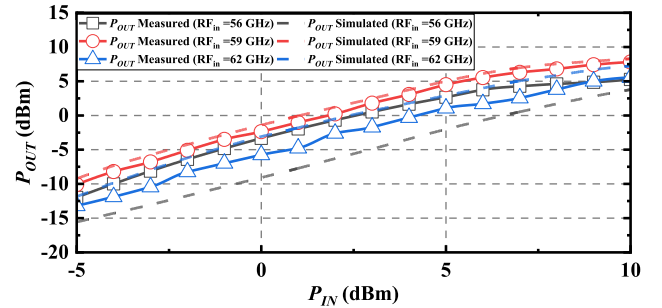


Fig. 22. Measured and simulated P_{OUT} versus P_{IN} .

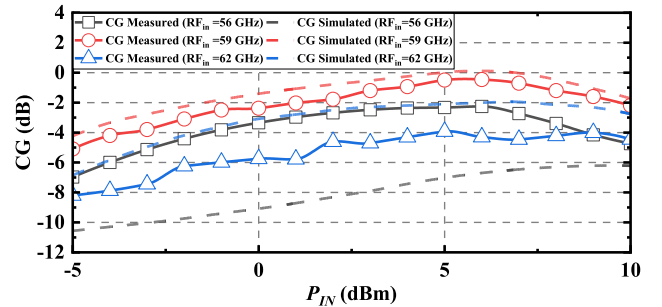


Fig. 23. Measured and simulated CG versus P_{IN} .

that the input power provided to the chip matches the intended value. Then, to get the accurate output power, the output chain is calibrated using 90–140-GHz network calibration, and the

TABLE I
COMPARISON OF STATE-OF-THE-ART D-BAND AND F-BAND FREQUENCY DOUBLERS

Ref	Tech.	Topology	Freq. (GHz)	P_{IN} (dBm)	P_{OUT} (dBm)	CG* (dB)	P_{DC} (mW)	V_{DD} (V)	Efficiency (%)	Area (mm ²)
MWCL'2014[15]	130-nm CMOS	PA+Dou.	134	15	4.2	-3.1	790	1.5	0.33	N/A
IMS'2014[16]	45-nm SOI CMOS	Dou.	135-160	7	3.5	-3	25	1	9.0	0.44
RFIC'2015[17]	32-nm SOI CMOS	Dou.+ Dou.	124-158	26.5	2.3	-21	32	1.1	5.3	0.29
EuMIC'2015[18]	28-nm CMOS	Dou.	105-130	6.2	4	-2.2	14	0.9	18	0.17
RFIT'2017[19]	65-nm CMOS	Dou.+Buffer	100-123	-8	5	17.2**	116	1	2.7	0.83
RFIC'2022[20]	22nm FD-SOI	Tri.+DA+Dou.+PA	127-162	1	5.1	6.85**	47	0.8	8.49	0.46
RFIC'2016[35]	90-nm SiGe	Dou.+Buffer	138-170	0.7	5.6	4.9**	36	1.5	10.9	0.48
TMTT'2021[36]	90-nm SiGe	Dou.	128-140	16/10	9.4/4	0.87	72	1.9	12.1	0.57
SSCL'2021[37]	55-nm SiGe	Buffer+Dou.	108-155	-6	8.1	13.1**	97.5	3	6.3	N/A
MWCL'2014[38]	250-nm InP DHBT	Buffer+Tri.	119-135	3	10	7**	45	1.6	12	1.5
TMTT'2014[39]	35-nm InGaAs mHEMT	Dou.	100-208	14	1.4	-7	14.4	0.8	9.6	0.375
This Work	55-nm CMOS	Dou.	112-125	10	7.8	-0.46	88	1.2	6.8	0.21

* Maximum value ** With buffer

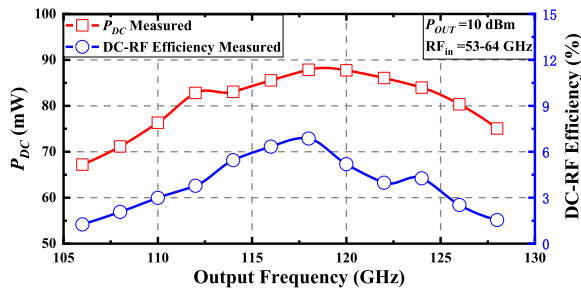


Fig. 24. Measured P_{dc} and efficiency versus RF_{in} at $P_{IN} = 10$ dBm.

IL of the output chain at 106–128 GHz is 4.9–3.7 dB. Note that the loss of WR8–WR10 Taper is ignored in the output chain, so the actual output power is slightly higher.

Fig. 21 depicts the output power P_{OUT} and output frequency RF_{out} under the two situations of 6- and 10-dBm input power P_{IN} . The 3-dB operating bandwidth obtained from the simulation is 113–126 GHz while the measurement value is 112–125 GHz, and the operating frequency of the measurement

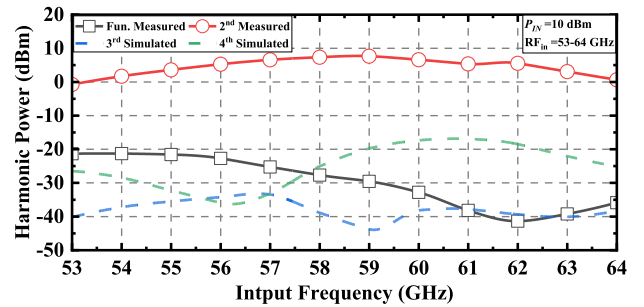


Fig. 25. Measured and Simulated harmonics of the proposed doubler versus RF_{in} at $P_{IN} = 10$ dBm.

value is relatively down 1 GHz, which generally agrees well with the simulation value in this frequency band.

Fig. 22 illustrates the measured and simulated P_{OUT} versus P_{IN} at 56/59/62 GHz. The class-F doubler achieves a saturation output power of 7.8 dBm when P_{IN} is 10 dBm. The measurement of the conversion gain (CG) is shown in Fig. 23, the measurement results coincided well with the simulation results, and when the input power is 6 dBm, the maximum CG achieved is -0.46 dB.

$|P4|_{\max}$ when $|V_{D1}|$ and $|V_{D3}|$ are at the large signal. Therefore, it is expected to maximize $P4$ for a maximum $P_{\text{out}}^{2f_0}$ and the simplified $P_{\text{out}}^{2f_0}$ is derived in (10). To maximize $P4$, (16), as shown at the bottom of the previous page, the optimal phase conditions; then, the optimal ϕ_{1-3} is shown in (11). Then, the optimal v_D shown in (12) shows that the peak of V_{D2} overlaps the valleys of V_{D1} and V_{D3} , which leads to a large triode region ($\theta_2 - \theta_1$) and is good for efficiency.

APPENDIX B DERIVATION OF EFFICIENCY

The Norton equivalent circuit of the FET based on EKV mode is shown in Fig. 26. $Y_{S,nf0}$ and $Y_{L,nf0}$ are denoted as the parasitic conductor of the FET and the load conductor, respectively, and the conductor seen from the drain terminal into FET is denoted as $Y_{D,nf0}$. Then, the 2nd-order harmonic load current is $i_{L2} = i_{D2} + i_{Y2}$, where i_{Y2} is the parasitic current. For the maximum 2nd-order harmonic power transmission, $Y_{L,2f0} = (Y_{D,2f0})^*$. Hence, the 2nd-order harmonic power generation is equal to the 2nd-order harmonic output power $P_{\text{out},2\text{nd}}$, as shown in (17). Based on the EKV model, the dc current of i_D is denoted as i_{D0} in (18). Then, the drain efficiency is derived in (19). For a 78- $\mu\text{m}/55\text{-nm}$ FET in 55-nm bulk CMOS technology, $|Y_{S,2f0}| = 24\text{ m}\Omega$ and $|Y_{S,0}| = 7\text{ m}\Omega$

$$\begin{aligned} P_{\text{out},2\text{nd}} &= P_{\text{out}}^{2f_0} = \frac{1}{2\pi} \int_0^{2\pi} (i_{D2} + i_{Y2}) \cdot V_{D2} \\ &= \frac{1}{2\pi} \int_0^{2\pi} (i_{D2} + |Y_{S,2f0}| \cdot V_{D2}) \cdot V_{D2} \end{aligned} \quad (17)$$

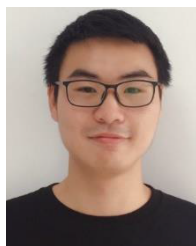
$$\begin{aligned} i_{D0} &= \frac{1}{2\pi} [|i_{D0}^A(\theta_1 + \pi - \theta_2)| + |i_{D0}^T(\theta_2 - \theta_1)|] \\ &= \left| \frac{1}{8\pi} \beta(\theta_1 + \pi - \theta_2) [|V_G|^2 + 2(V_0 - V_{\text{th}})^2] \right| \\ &\quad + \left| \frac{1}{8\pi} \beta(\theta_2 - \theta_1) \left[2|V_G||V_{D1}| \cos(\phi_1) + 4V_{\text{DD}}(V_0 - V_{\text{th}}) \right. \right. \\ &\quad \left. \left. - |V_{D1}|^2 - |V_{D2}|^2 - |V_{D3}|^2 - 2V_{\text{DD}}^2 \right] \right| \end{aligned} \quad (18)$$

$$\begin{aligned} \eta &= \frac{P_{\text{out},2\text{nd}}}{P_{\text{DC}}} = \frac{P_{\text{out},2\text{nd}}}{(i_{D0} + i_{Y0}) \cdot V_{\text{DD}}} \times 100\% \\ &= \frac{P_{\text{out},2\text{nd}}}{(i_{D0} + Y_{S,0} \cdot V_{\text{DD}}) \cdot V_{\text{DD}}} \times 100\%. \end{aligned} \quad (19)$$

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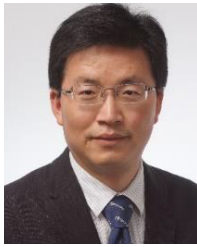
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